

EduVision – Dashboard Testing Report

Module 7 – Testing & Validation

Purpose: Document the testing performed on the four EduVision Tableau dashboards, including KPI accuracy, ranking logic, required educational metrics, data quality, and dashboard interactivity.

Project	EduVision – University Analytics Dashboard
Testing phase	Module 7 – Testing & Validation
Dataset	university_final_dataset.xlsx
Tableau workbook	EduVision_DV.twbx
Overall status	PASS – Module 7 complete

1. Test Scope

Testing covered University Overview, Research Analytics, Student Analytics, and Country Comparison. Checks included KPI calculations, rankings, educational metrics, data quality, visualization structure, and live filter/action behavior.

2. Test Cases

ID	Test Area	Expected / Validation	Result
TC-01	Total Universities KPI	COUNTD(University Name) counts unique universities.	PASS
TC-02	Academic Reputation KPI	Uses AVG(Academic Reputation Score).	PASS
TC-03	Female/Male Percentage	Percentage fields are aggregated on the intended percentage scale.	PASS
TC-04	International Student KPI	Matches the underlying international student ratio.	PASS
TC-05	Faculty-to-Student KPI	Calculated consistently from Faculty Count / Total Students.	PASS
TC-06	Best University Rank	Uses minimum world rank by country/year.	PASS
TC-07	Country Average Rank	Uses country-level average rank; lower rank is better.	PASS
TC-08	Ranked University Count	Consistent with final ranking logic.	PASS
TC-09	Research Productivity Trend	Required yearly research trend is	PASS

		present with intended aggregation.	
TC-10	Regional Education Trend	Required regional trend is present across years.	PASS
TC-11	Faculty-to-Student Ratio Analysis	Required analysis is present.	PASS
TC-12	Student Enrollment Across Universities	Required enrollment comparison is present.	PASS
TC-13	Country Ranking Comparison	Required country ranking comparison is present.	PASS
TC-14	Filter Actions	Filter1 and Filter2 are present.	PASS
TC-15	Live Dashboard Interaction	Filters/interactions were manually tested and worked.	PASS
TC-16	Orphaned Worksheets	Redundant unused worksheets were removed from final workbook.	PASS

3. Dashboard Results

Dashboard	Tested Components	Result
University Overview	KPIs, university comparisons, country view, university type and score metrics.	PASS
Research Analytics	Research productivity trend, citations, publications, H-index and research comparisons.	PASS
Student Analytics	Student totals, enrollment, faculty-to-student ratio, international students and gender.	PASS
Country Comparison	Regional education trend, country ranking and country performance comparisons.	PASS

4. Issues Found and Corrections

Issue	Finding	Correction	Status
Total Universities showed 10	An unintended Top-N filter affected the KPI.	Removed the unintended filter and retained COUNTD.	Resolved
Academic Reputation used wrong measure	Research Impact Score appeared	Changed to AVG(Academic	Resolved

	under the Academic Reputation label.	Reputation Score).	
Female/Male values were inflated	Percentage fields were summed across records.	Changed aggregation to average percentage values.	Resolved
Faculty-to-student KPI inconsistency	Earlier KPI timing differed from final imputed fields.	Recalculated from final faculty and student fields.	Resolved
Ranking discrepancies	Some rank fields reflected earlier calculation timing.	Recalculated consistently with final cleaned data.	Resolved
Orphaned worksheets	Three unused worksheets were not on dashboards.	Removed redundant worksheets.	Resolved

5. Data & Methodology Notes

The final dataset contains 21,127 records, covers 2017–2026, and contains 3,605 unique universities. The 668 authoritative 2024 records identified in Method_Notes were retained. The 2026 records were retained as part of the extended trend dataset and documented rather than silently removed.

The pronounced 2024 Research Productivity variation was retained because it is a characteristic of the supplied dataset; the trend was not artificially smoothed or deleted.

6. Manual Interaction Testing

Filter1 and Filter2 were tested manually in Tableau across the four dashboards. Filters responded correctly and did not produce unexpected blank results during the test.

7. Final Result

FINAL RESULT: PASS

All Module 7 testing areas have been addressed. Calculation issues were corrected, required dashboard analyses are present, redundant worksheets were removed, methodology notes were documented, and dashboard interactions were manually verified.

Module 7 – Testing & Validation: COMPLETE